A Non-Monotonic Velocity-Field Formal Solver for CMFGEN: Numerical Approach, Source-Code Modifications, Benefits, and Limitations

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Abstract

CMFGEN solves the comoving-frame radiative transfer equation under the assumption of monotonic outward expansion, parameterised by the Sobolev gradient variable $\sigma \equiv \mathrm{d}\ln v/\mathrm{d}\ln r - 1 \geq -1$. Hydrodynamic snapshots from STELLA of interacting supernovae—containing shocks, cold dense shells, and velocity reversals—routinely produce $\sigma \ll -1$, causing the legacy short-characteristics solver `FG_J_CMF_V13` to generate negative optical depths, NaN cascades, and segmentation faults. We describe an opt-in non-monotonic formal solver, `FG_J_CMF_NONMONO_V1`, that separates the Sobolev optical-depth *magnitude* from the flow *direction*, enabling piecewise short-characteristics integration across multiple resonance crossings. We further describe a critical refinement: threading the true unclamped σ into the formal solver via a parallel `SIGMA_TRUE` array, while retaining the clamped σ for all infrastructure routines (moment equations, Ng acceleration, Sobolev escape probabilities, grid regridding). We assess the benefits (correct Sobolev optical depths in shock zones, restored multi-root resonance detection) and the remaining limitations (low optical depth of test snapshots, `DC_METH=R` temperature smearing, LAMBDA-iteration capping).

Contents

1 Introduction

1.1 Physical context

This work arises from the modelling of SN 2024pba, a two-event interacting-supernova scenario in which a later explosion (SN B) drives a shock into the metal-rich ejecta of an earlier Type IIb explosion (SN A). The scientific objective is to determine whether shock heating and ionisation of the SN A ejecta can reproduce the observed spectrum, particularly strong absorption associated with H, He, O, Si, Ca, and Fe.

The modelling workflow couples the radiation-hydrodynamics code STELLA to the non-LTE radiative-transfer code CMFGEN. STELLA produces hydrodynamic snapshots (radius, velocity, density, temperature, composition as functions of mass coordinate and time) that are converted into the CMFGEN input file `SN_HYDRO_DATA`. CMFGEN then solves the statistical-equilibrium and comoving-frame transfer equations to produce a synthetic spectrum.

1.2 The monotonic-expansion assumption

CMFGEN was designed for homologously expanding supernova atmospheres and stellar winds, in which velocity increases monotonically outward. The key structural variable is the Sobolev gradient

$$\sigma \equiv \frac{\mathrm{d}\ln v}{\mathrm{d}\ln r} - 1, \tag{1}$$

which satisfies $\sigma \geq -1$ for all monotonic expansion profiles. The angle-dependent Sobolev factor entering the comoving-frame transfer equation is

$$T_2(\mu) = 1 + \sigma\,\mu^2, \tag{2}$$

where $\mu = \cos\theta$ is the ray angle cosine relative to the radial direction. When $\sigma \geq -1$, $T_2 > 0$ for all μ, and the comoving frequency

$$\nu_{\mathrm{cm}}(r) = \nu_{\mathrm{obs}}\left(1 - \frac{v(r)}{c}\right) \tag{3}$$

crosses a given rest frequency at most once along any ray. The short-characteristics integration proceeds along monotonic sub-intervals with non-negative optical-depth increments.

1.3 The non-monotonic problem

Interacting supernovae violate this assumption. Shocks, cold dense shells, and reverse shocks produce velocity reversals in which $v(r)$ decreases outward over some region, giving $\sigma < -1$. In the STELLA snapshot of SN 2024pba at 35.57 days, 56 of 251 zones (22.3%) have $\sigma < -1$, with a minimum value of $\sigma = -94.66$.

When $\sigma < -1$, T_2 can change sign along a ray: $T_2 > 0$ for nearly tangential rays ($\mu \to 0$) but $T_2 < 0$ for nearly radial rays ($\mu \to 1$). The comoving frequency then crosses a given rest frequency *multiple times* along a single ray—a multi-root resonance. The legacy solver, which assumes a single resonance per ray, produces negative optical depths ($\mathrm{d}\tau \propto T_2 < 0$), negative intensities, and ultimately NaN cascades that crash the free-free gaunt-factor table lookup in `gff_vec`.

2 Numerical approach of the non-monotonic solver

2.1 Separation of magnitude and direction

The central design principle of `FG_J_CMF_NONMONO_V1` is to separate the Sobolev optical-depth *magnitude* from the flow *direction*. For each ray point i along impact parameter p:

$$\mu_i = \frac{z_i}{r_i}, \tag{4}$$

$$T_1 = \frac{v_i/c}{r_i}, \tag{5}$$

$$T_2 = 1 + \sigma_i\,\mu_i^2, \tag{6}$$

$$\Gamma_i = T_1\,T_2 \quad \text{(signed)}, \tag{7}$$

$$\Gamma_{\mathrm{mag},i} = T_1\ \max(|T_2|,\,10^{-12}) \quad (\geq 0), \tag{8}$$

$$s_i = \mathrm{sgn}(T_2) \quad (+1 \text{ expanding},\ -1 \text{ compressing}). \tag{9}$$

The non-negative magnitude Γ_{mag} is used wherever a strictly non-negative optical-depth increment is required: the differential optical depth $\mathrm{d}\tau$, and the Q term entering the extinction coefficient χ_{ray}. This guarantees $\mathrm{d}\tau \geq 0$ even in compression zones, eliminating the negative-optical-depth instability.

The signed Γ is retained for the source-function advection term $Q\,I_{\mathrm{prev}}$, where the sign carries the physical flow direction. The direction flag s_i records whether the flow is expanding $(+1)$ or compressing (-1) at each ray point.

The floor $\max(|T_2|, 10^{-12})$ prevents division by zero at resonance turning points (where $T_2 = 0$) without distorting the magnitude in the common case $|T_2| \gg 10^{-12}$.

2.2 Multi-root resonance scan

For a non-monotonic $v(r)$, the solver scans each ray for zones $[i, i{+}1]$ where $T_2(i)\,T_2(i{+}1) \leq 0$. These are resonance turning points—locations where the comoving frequency reaches an extremum and reverses direction relative to a given rest frequency.

No special interpolation is performed at the turning point. Instead, the piecewise short-characteristics integration naturally sums the source-function contribution over each monotonic sub-interval, because $\mathrm{d}\tau$ (built from Γ_{mag}) accumulates monotonically across each sub-interval, while the signed Γ and s_i preserve the advection direction.

In the test model, inner rays (small impact parameter) exhibit up to 8 turning points; outer rays exhibit 2–4. This is consistent with the shock structure in the STELLA snapshot.

2.3 Short-characteristics integration

The integration follows the same short-characteristics approach as the legacy solver:

1. Interpolate η, χ, χ_{scat} onto the ray grid.
2. Compute $\mathrm{d}\tau(i,p)$ from χ_{ray} and $\mathrm{d}z$, using Γ_{mag} for the Sobolev contribution.
3. Compute exponential integrals E_0, E_1, E_2, E_3 from $\mathrm{d}\tau$, with $\exp(-\mathrm{d}\tau)$ guarded against overflow ($\mathrm{d}\tau < 700$).
4. Solve the tridiagonal system for the radiation field $A_V(i,p)$.
5. Compute $J_\nu(i)$ from angle-weighted quadrature over all rays.

The non-monotonic modifications are confined to the $\mathrm{d}\tau$ and Q terms; the tridiagonal solve and quadrature are unchanged.

2.4 Negative-intensity guard

After the formal solution, the solver checks for negative mean intensities $J_\nu < 0$. If detected, it writes a diagnostic file `FG_J_CMF_NONMONO_V1_ERRORS` containing the frequency, J_ν, η, χ, and χ_{scat} at all depths, then halts. This is stricter than the legacy solver, which only checks the innermost point $R_J(1)$ for blow-up.

3 Infrastructure hardening

3.1 Design principle: clamp in infrastructure, not in the solver

The formal solver is designed to handle true negative σ. However, the rest of CMFGEN's infrastructure—moment equations, the BA matrix, Sobolev escape probabilities, grid regridding, Ng acceleration—was not designed for $\sigma < -1$ and produces NaN or unphysical results if given unclamped values.

The adopted principle is:

> *SIGMA clamps in infrastructure routines are always applied, regardless of* `NON_MONO_SOLVER`*. Only the formal solver* `FG_J_CMF_NONMONO_V1` *sees the true negative* σ*.*

Table **??** summarises the clamping at each site.

Table 1: SIGMA clamping in CMFGEN infrastructure.

Routine	Clamp	Reason
`rv_sn_model_sniin.f`	$\sigma \geq -0.999$	Hydro model setup
`extend_vtsigma.f`	$\sigma \geq -0.9999$	Grid extension
`adjust_r_grid_v4.f`	$\sigma \geq -0.999$ (smooth)	Regridding
`mom_j_cmf_v11.f`	$\sigma \geq -0.999$	Moment equations
`do_ng_accel_band_v4.f`	$\sigma \geq -0.999$	Ng acceleration
`sobjbar_sim.f`	Dessart+2015 guard	Sobolev escape probability
`fg_j_cmf_nonmono_v1.f`	**No clamp**	**Formal solver**

3.2 LAMBDA-iteration capping

The LAMBDA iteration in `SOLVEBA_V13` uses a `LIMIT` option that caps population corrections. The original code only capped decreases (STEQ $> 1.1 \rightarrow 0.999$) but not increases. With non-monotonic velocity fields, catastrophic population increases (up to 10^{62}%) occurred at the outer boundary, corrupting T and crashing `gff_vec`.

The fix caps both directions:

```
IF (STEQ(J,I) > 1.1)   STEQ(J,I) =  0.999    ! cap decreases
IF (STEQ(J,I) < -1.1)  STEQ(J,I) = -0.999    ! cap increases
```

This reduced the population explosion from 10^{62}% to ~ 110% and eliminated the NaN cascade.

3.3 Free-free gaunt-factor NaN guard

The `gff_vec` routine computes $\log T$ and uses it to index a pre-computed table. If T becomes NaN, the table access segfaults. The guard:

```
IF (.NOT. (LOGT > 0.0 .OR. LOGT < 0.0)) LOGT = 3.6   ! NaN check
IF (LOGT < 2.0) LOGT = 2.0                            ! floor (100 K
   )
IF (LOGT > 6.0) LOGT = 6.0                            ! cap (10^6 K)
```

converts NaN to a safe default and clamps extreme values to the table range.

4 The SIGMA_TRUE threading: true unclamped σ to the formal solver

4.1 The problem

The original implementation clamped σ to ≥ -0.999 at hydro read-in (`rv_sn_model_sniin.f`) *before* the formal solver ever saw it. This meant `FG_J_CMF_NONMONO_V1` received $\sigma = -0.999$ even when the true hydrodynamic value was $\sigma = -94.66$.

The consequence is severe. The Sobolev line optical depth scales as

$$\tau_{\rm Sob} \propto \frac{v/c}{r} |T_2| = T_1 \, |1 + \sigma\mu^2|. \tag{10}$$

For a radial ray ($\mu = 1$) through a zone with true $\sigma = -94.66$:

- True $|T_2| = |1 - 94.66| = 93.66$
- Clamped $|T_2| = |1 - 0.999| = 0.001$
- Optical-depth error: $\sim 9 \times 10^4$ too small.

Lines forming in the shock-compressed metal-rich ejecta—the scientifically critical H, He, O, Si, Ca, Fe absorption features—had severely underestimated Sobolev optical depths on the radial rays that dominate the P-Cygni absorption troughs.

4.2 The solution: a parallel SIGMA_TRUE array

A parallel `SIGMA_TRUE` array carries the unclamped value, threaded into the formal solver only. The clamped `SIGMA` is retained for all infrastructure (Table **??**).

4.2.1 Data flow

1. `SIGMA_TRUE(ND)` is declared in `MOD_CMFGEN` and allocated alongside `SIGMA` in `cmfgen.f`.
2. In `rv_sn_model_sniin.f`, the true value is saved *before* the clamp:

```
SIGMA_TRUE(1:ND) = SIGMA(1:ND)      ! save unclamped
DO I=1,ND
  IF(SIGMA(I) .LT. -0.999_LDP) SIGMA(I) = -0.999_LDP
END DO
```

4.2.2 Grid extension

The legacy extension routine `EXTEND_VTSIGMA` interpolates in $\log(\sigma+1)$ space, which is undefined for $\sigma < -1$. A new routine `EXTEND_SIGMA_TRUE` was created that applies the cubic interpolation coefficients (computed by `REXT_COEF_V2` in the $\log r$ plane) *linearly* to σ:

$$\sigma_{\rm true,ext}(i) = \sum_{j=0}^{3} C_{j,i} \, \sigma_{\rm true}(j + {\rm INDX}(i)) \, . \tag{11}$$

This is well-defined for any σ value. No clamping is applied. The clamped `SIGMAEXT` is still produced by `EXTEND_VTSIGMA` for infrastructure use; the two extensions coexist.

`EXTEND_SIGMA_TRUE` is called after each of the four `EXTEND_VTSIGMA` call sites in `cmfgen_sub.f`, producing `SIGMA_TRUE_EXT` on the refined grid.

4.2.3 Single chokepoint in the formal solver

The formal solver `FG_J_CMF_NONMONO_V1` receives `SIGMA_TRUE_IN` as a new dummy argument. At a single chokepoint (line 608), the main-grid portion of the extended array is filled from the true value:

```
SIGMA_EXT(ND_ADD+1:ND_EXT) = SIGMA_TRUE_IN(1:ND)
```

All downstream quantities are derived from `SIGMA_EXT` and therefore automatically use the true values:

- `SIGMA_COEF` (cubic interpolation coefficients, built from `SIGMA_EXT`)
- `SIGMA_RAY` (ray-grid values, interpolated from `SIGMA_COEF`)
- Γ, T_2, $\Gamma_{\rm mag}$, s_i (Eqs. **??**)
- Resonance turning-point scan (T_2 sign changes)
- $\mathrm{d}\Gamma/\mathrm{d}r$ derivatives

The outer β-law extension (`THK` block) still uses the clamped `SIGMA(1)`. This is intentional: the outer extension is monotonic, so $\sigma(1) = \sigma_{\rm true}(1)$ there and the clamp has no effect.

4.2.4 Call-site routing

In `comp_j_blank.f`, the two call paths are:

- **ACC path** (acceleration/extended grid): passes `SIGMA_TRUE_EXT`.
- **Main path** (frequency iteration, produces the final spectrum): passes `SIGMA_TRUE`.

When `NON_MONO_SOLVER = .FALSE.`, the legacy `FG_J_CMF_V13` is called with its original (untouched) signature; `SIGMA_TRUE` is populated but never read, so legacy behaviour is bit-for-bit unchanged.

4.2.5 OBS-stage independence

The observed-spectrum stage uses `COMP_OBS_V3` and `OBS_FRAME_SUB_V9`, which transform the stored $I^+(p)$ (computed by the formal solver with true σ) to the observer frame using velocity v only. Neither routine references σ. The observed spectrum therefore inherits the correct comoving-frame radiation field without requiring further modification.

5 Benefits

5.1 Correct Sobolev optical depths in shock zones

The primary benefit. With the true σ threaded to the formal solver, Sobolev line optical depths in compression zones are now physically correct. For the worst zones in the test model ($\sigma_{\rm true} = -94.66$), the optical depth on radial rays increases by a factor of $\sim 9 \times 10^4$ relative to the clamped value. P-Cygni absorption troughs—formed by radial rays through the line-of-sight column—now have correct depths.

5.2 Restored multi-root resonance detection

The resonance turning-point scan (T_2 sign changes) now operates on the true σ. With the clamped $\sigma = -0.999$, $T_2 = 1 - 0.999\mu^2$ only crosses zero at $\mu \to 1$ and only barely; the dramatic multi-resonance structure of the true velocity reversal was flattened. With the true σ, the solver correctly detects up to 8 turning points on inner rays, matching the shock structure in the STELLA snapshot.

5.3 Stable iteration without crashes

The infrastructure hardening (LAMBDA capping, gaunt-factor NaN guard) eliminates the $10^{62}\%$ population explosions and segmentation faults that crashed the legacy solver on non-monotonic models. The test model iterates stably through 30+ iterations with no NaN, no SIGSEGV, and no negative intensities.

5.4 Bit-for-bit legacy preservation

When `NON_MONO_SOLVER = .FALSE.` (the default), the legacy solver `FG_J_CMF_V13` is called with its original signature. The `SIGMA_TRUE` array is populated but never read. All infrastructure clamps are unconditional. Legacy models produce bit-for-bit identical output to the baseline—verified by source inspection of the routing logic in `comp_j_blank.f`.

6 Drawbacks and limitations

6.1 Low optical depth of the test snapshot

The test snapshot (35.57 days) has a Rosseland optical depth of only $\tau_{\rm ross} = 1.84$ at the inner boundary, far below the $\tau \gtrsim 50$ required for the diffusion approximation. The continuum luminosity converges to $\sim 6.6 \times 10^9\, L_\odot$ versus $L_\star = 5.4 \times 10^8\, L_\odot$ from the STELLA header—an order-of-magnitude mismatch. The absolute flux scale and continuum shape of the spectrum are therefore unreliable for this epoch.

This is a property of the snapshot epoch, not of the solver. Earlier snapshots or more massive models would have higher τ. However, for the 35.57 day test, no spectral output can be trusted quantitatively.

6.2 DC_METH=R temperature smearing

Non-monotonic temperature structure requires `DC_METH=R` (radius interpolation) instead of `DC_METH=TR` (temperature interpolation), because `TR` crashes `REGRID_LOG_DC_V1` on non-monotonic T.

In shock-cooling regions with steep temperature gradients, radius-based interpolation does not preserve the temperature structure as accurately as temperature-based interpolation. This smears ionisation boundaries (e.g. O I/II/III, Fe II/III/IV) across a radius interval, shifting which ion dominates and changing line strengths. The effect compounds with the Sobolev optical-depth correction (§**??**) in the same shock-cooling zones.

6.3 LAMBDA-iteration capping artifacts

The `LIMIT/MAX_LAM=10`^{2} *capping in* `SOLVEBA_V13` *damps population corrections. The diagnostic output reports thous*
−−it may converge to a solution that is stable but not the true statistical−equilibrium solution. Line source functions

6.4 Negative-intensity clamping

`Negative intensities (`$I_M < 0$, $I_P < 0$`) in compression zones are clamped to` 10^{-30}`.`
`This is a safety net. The fact that negatives appear means the source function is`
`being mis-evaluated there; clamping to a tiny positive value suppresses whatever`
`emission or absorption feature would have formed, so those regions contribute approximately`
`zero to the spectrum---an unphysical blackout in the worst compression zones.`

6.5 No completed spectral synthesis

`The test model was run on a 2-hour interactive partition and did not reach the OBS`
`(spectral synthesis) stage before the wall limit. No CMFGEN spectrum output exists`
`for the non-monotonic model. Every claim about the actual spectral appearance is`
`supported by physics but not verified by output. This is a resource limitation,`
`not a physics one.`

6.6 No monotonic-model regression test

The non-monotonic solver has not been tested on a standard monotonic supernova model to confirm it produces identical results to FG_J_CMF_V13 when $\sigma \geq -1$ everywhere. While the algorithm reduces to the standard Sobolev integration in that limit ($T_2 > 0$, $\Gamma_{\rm mag} = \Gamma$, $s_i = +1$), this has not been demonstrated numerically.

6.7 Physical correctness not established

Numerical convergence does not establish physical correctness. The low optical depth, temperature smearing, and population capping all introduce systematic errors whose net effect on the spectrum has not been quantified. A spectrum that appears to match observations could be matching for the wrong physical reasons.

7 Configuration

7.1 VADAT settings

Required for non-monotonic models:

```
T   [NON_MONO_SOLVER]      ! enable the non-monotonic formal solver
R   [DC_METH]              ! radius interpolation (TR crashes on non
   -monotonic T)
```

Recommended:

```
LIMIT [LAM_SCALE_OPT]     ! cap LAMBDA population changes
1.0D2 [MAX_LAM]           ! allow large LAMBDA steps (default 1.2
   too conservative)
```

7.2 Executable

The non-monotonic solver is compiled into cmfgen_dev.exe in the cmfgen_nonmono_dev source tree:

```
cd cmfgen_nonmono_dev && make
```

The slurm script must set:

```
export cmfdist="/path/to/cmfgen_nonmono_dev"
```

7.3 YAML configuration

```
cmfgen:
  vadat_overrides:
    NON_MONO_SOLVER: T
    DC_METH: R
    LAM_SCALE_OPT: LIMIT
    MAX_LAM: 1.0D2
```

8 Summary

The non-monotonic formal solver FG_J_CMF_NONMONO_V1 extends CMFGEN to handle hydrodynamic snapshots with shocks, cold dense shells, and velocity reversals ($\sigma < -1$) that crash the legacy solver. The key algorithmic innovation is the separation of the Sobolev optical-depth magnitude ($\Gamma_{\rm mag} \geq 0$) from the flow direction ($s_i = \pm 1$), enabling piecewise short-characteristics integration across multiple resonance crossings without producing negative optical depths.

The SIGMA_TRUE threading refinement ensures the formal solver receives the true unclamped σ from the hydrodynamic model, while the clamped σ protects the infrastructure. This restores physically correct Sobolev optical depths in shock zones---the dominant correction---and enables proper multi-root resonance detection.

The remaining limitations (low optical depth of the test snapshot, DC_METH=R temperature smearing, LAMBDA capping) are secondary and do not invalidate the solver design. Full validation requires a model run to spectral synthesis on a snapshot with sufficient optical depth ($\tau \gtrsim 50$), which is left as future work.

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